

DriftTrace Background and Reproducibility Critique

Suggested text for Related Work section

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Journal Paper:

From Detection to Decision Support: Role-Aware Human-Centred Explanations of AI-based Intrusion Detection Systems for SOC Under Concept Drift

Full Version — Related Work Section

Among the most relevant recent contributions to explainable concept drift detection in cybersecurity is DriftTrace [CITE], which provides both a detection mechanism and a sample-level explanation module for identifying drifting network flows in intrusion detection systems. DriftTrace operates through three sequential modules. First, a contrastive autoencoder maps input network flows into a low-dimensional latent space, where same-class samples are pulled together and different-class samples are pushed apart via a pairwise contrastive objective. Second, an IQR-based detection algorithm determines whether each incoming test sample lies within the known training distribution by comparing its latent distance to each known class centroid against empirically computed quartile bounds. Third, a greedy feature selection algorithm identifies the specific input features most responsible for the detected drift, by iteratively replacing features in the drift sample with values from a representative known-class sample and measuring the resulting reduction in latent distance. This three-stage pipeline is notable for operating at the sample level rather than the distribution level, enabling fine-grained per-flow explanations without requiring analyst labelling of drift samples.

DriftTrace is evaluated on the CSE-CIC-IDS2018 network intrusion dataset, reporting a mean detection F1 of 0.97 +/- 0.02, substantially above both the Vanilla Autoencoder baseline (F1 = 0.84 +/- 0.04) and the TRANSCENDENT conformal prediction baseline (F1 = 0.75 +/- 0.04). On the explanation side, DriftTrace achieves a fidelity distance of 0.0325 and a boundary crossing rate of 83.65% for the Infiltration hold-out scenario, indicating that in over four-fifths of detected drift cases, the selected explanation features are sufficient to reclassify the drift sample as a known class after feature replacement.

Despite these promising results, DriftTrace carries several limitations relevant to the present work. First, the explanation produced by Algorithm 2 consists of a ranked list of input feature indices selected by a greedy distance-reduction criterion. While this identifies which features are statistically responsible for the drift, it does not translate these features into actionable intelligence for a Security Operations Centre (SOC) analyst. A feature value such as "Flow Bytes/s = 2.4e6" conveys little operational meaning without statistical context, comparison to known baselines, or natural language framing. The paper itself designates the connection between detection and analyst-facing explanation as future work. Second, DriftTrace's explanation model generates a single feature mask for each drift sample without accounting for the role or technical background of the receiving analyst. A network engineer and a security manager both working within the same SOC require fundamentally different levels of abstraction when assessing

the same alert. Third, DriftTrace assumes immediate label availability for identified drift samples during the adaptation phase, whereas in practice SOC workflows involve triage delays and asynchronous labelling.

The present work adopts DriftTrace's detection and explanation modules as its foundation while extending it with a role-aware natural language explanation layer powered by a large language model (LLM). Rather than outputting a feature mask, our system converts the statistical outputs of the detection pipeline into structured, contextualised natural language that adapts its vocabulary, granularity, and recommended actions to the inferred role of the receiving analyst.

It is important to note that our implementation of DriftTrace's detection component was subjected to rigorous independent reproduction before being adopted as a base. Across nineteen experiment versions (fourteen unique configurations) implementing every documented and plausibly inferred hyperparameter of DriftTrace on the same IDS2018 dataset, our best reproduction achieved an average detection F1 of 0.842 with a boundary crossing rate of 40.5% for the Infiltration scenario — both below the paper's claims of 0.97 and 83.65% respectively. Detailed analysis revealed that the gap is attributable to an undocumented positive-pair sampling ratio in the contrastive training loop, which controls the width of same-class latent clusters and therefore the calibration of IQR detection bounds. When trained with our implementation, same-class clusters converge to a Q3 distance of approximately 0.003-0.010 from the centroid, producing IQR upper bounds of 0.005-0.018. For the paper's 83.65% boundary crossing to be achievable, clusters would need Q3 approximately 0.030-0.060 — approximately ten times wider — which was not achievable under any documented hyperparameter combination within the 300-epoch training budget. A query to the corresponding author (Dr. Lixin Zhao, zhaolixin@iie.ac.cn) seeking clarification received no response at the time of writing. The complete code, results, and analysis for all nineteen experiment scripts are publicly available at <https://github.com/MNadeemYousaf>, providing full transparency for the detection foundation on which the present work builds. Despite the reproducibility gap, DriftTrace's methodology is sound and our reproduction consistently surpasses both paper-reported baselines (Vanilla AE F1=0.84, TRANSCENDENT F1=0.75), validating the detection component as a viable foundation for the human-centred explanation layer proposed in this work.

Condensed Version (for word-limited journals)

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Submission Notes

- Replace [CITE] with the DriftTrace IEEE TIFS 2026 BibTeX key.
- Replace <https://github.com/MNadeemYousaf> with the specific repository URL.
- Full version recommended — transparency in reproducibility is increasingly expected.
- The condensed 4-sentence version suits journals with strict word limits.
- +/- is used here in place of the $\pm$ symbol for PDF font compatibility.